Final Project Report: Bank Marketing Campaign Analysis

1. Executive Summary

- **Objective:** To identify key drivers of term deposit subscriptions and optimize marketing outreach.
- **The "Big Number":** Out of 45,211 leads, the campaign achieved a **11.70% conversion rate**.
- **Primary Recommendation:** Pivot marketing spend toward Student and Retired demographics while implementing a "4-minute engagement" target for sales representatives.

2. The Conversion Funnel (Process Insight)

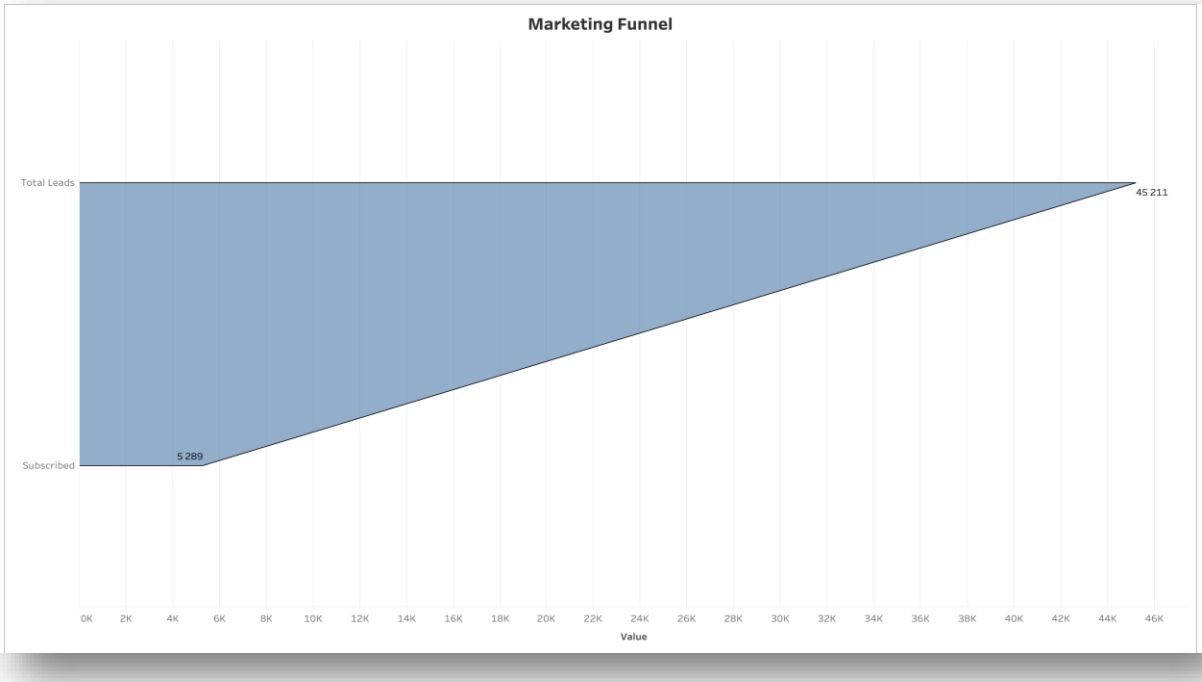


Figure 1 Note: Out of 45,2111 initial leads, the campaign achieved 5,289 successful subscriptions, representing an overall conversion rate of 11.70%.

- **Analysis:** We see a significant drop-off from initial contact to subscription.
- **Key Finding:** Only 1 out of every 8.5 leads converts. This high volume of "No" responses suggests the need for better lead pre-qualification.

3. High-Value Customer Segments (The "Who")

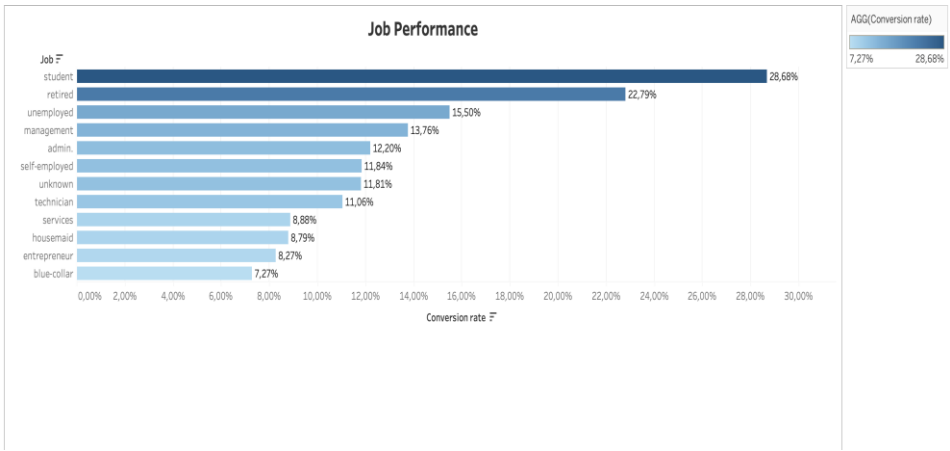
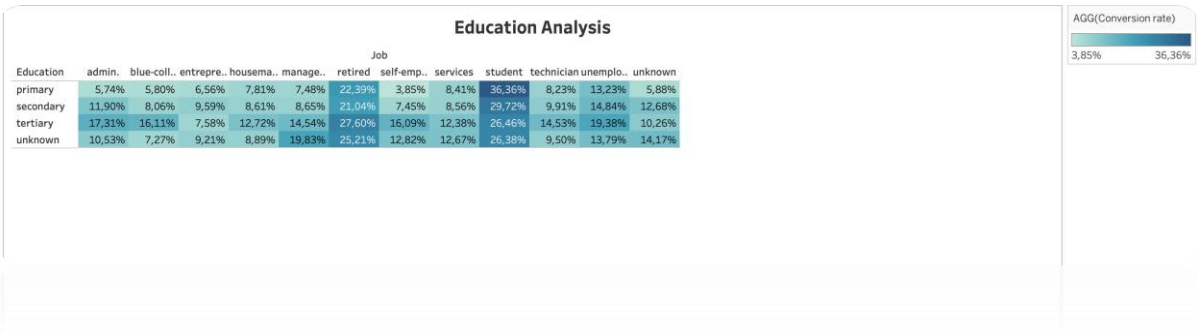


Figure 2 Note: Students and Retired individuals emerge as the highest-performing segments, converting at more than double the rate of “Blue-collar" workers.



Education Analysis

Education	Job: admin.	blue-coll..	entrepre..	housema..	manage..	retired	self-emp..	services	student	technician	unemplo..	unknown
primary	5,74%	5,80%	6,56%	7,81%	7,48%	22,39%	3,85%	8,41%	36,36%	8,23%	13,23%	5,88%
secondary	11,90%	8,06%	9,59%	8,61%	8,65%	21,04%	7,45%	8,56%	29,72%	9,91%	14,84%	12,68%
tertiary	17,31%	16,11%	7,58%	12,72%	14,54%	27,60%	16,09%	12,38%	26,46%	14,53%	19,38%	10,26%
unknown	10,53%	7,27%	9,21%	8,89%	19,83%	25,21%	12,82%	12,67%	26,38%	9,50%	13,79%	14,17%

AGG(Conversion rate)
3,85% 36,36%

Figure 3 Note: Tertiary-educated leads consistently show higher subscription rates across most job categories, with significant performance peak in the “Student-Secondary” and “retired-Tertiary” intersections.

- **Top Performers: Students (28.68%)** and **Retired (22.79%)** individuals convert at more than double the baseline rate.
- **The "Education Lift":** Tertiary-educated individuals consistently show higher engagement across almost all job categories.
- **Strategic Move:** Lower the priority for "Blue-collar" segments (7.27% conversion) and reallocate those resources to the Student/Retired outreach programs.

4. Behavioral Drivers (The "How")

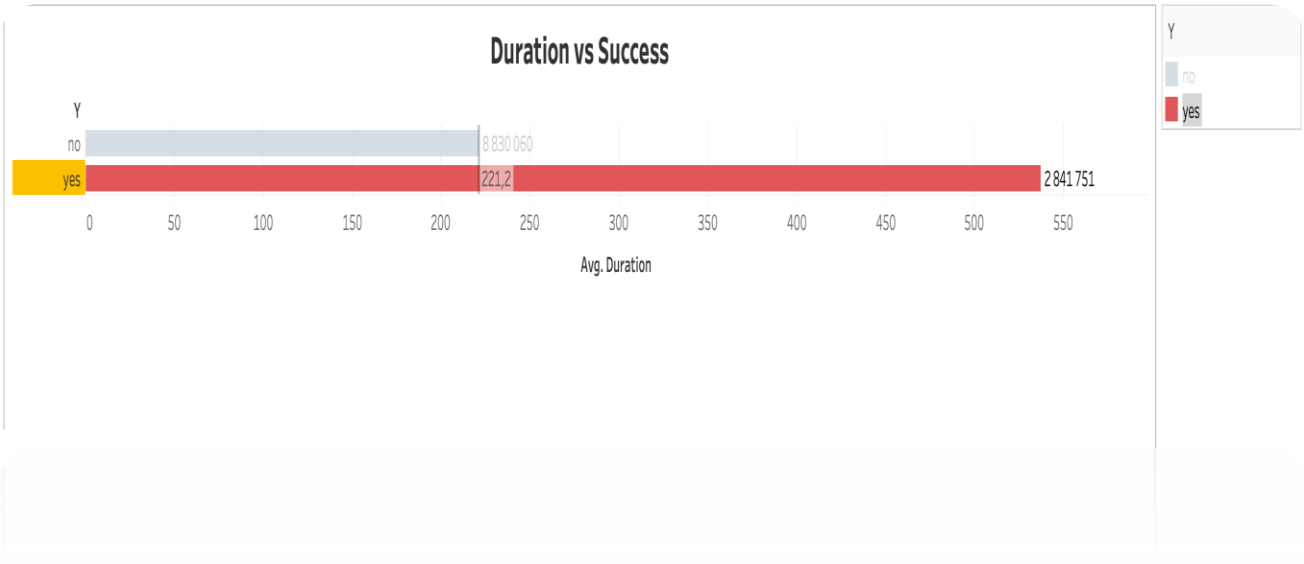


Figure 4 Note: Successful conversions are strongly correlated with higher engagement time, averaging 537 sec (8.9 mins) compared to only 221 sec (3.7 mins) for unsuccessful leads.

- **The 4-Minute Threshold:** Failed calls average **3.7 minutes**, while successful conversions average **8.9 minutes**.
- **The Insight:** There is a "dead zone" around the 4-minute mark. If a representative hasn't secured interest by this point, the likelihood of conversion plummets.
- **Recommendation:** Sales training should focus on "The Hook"—securing a commitment to keep talking past the first 240 seconds.

5. Seasonal Opportunities (The "When")

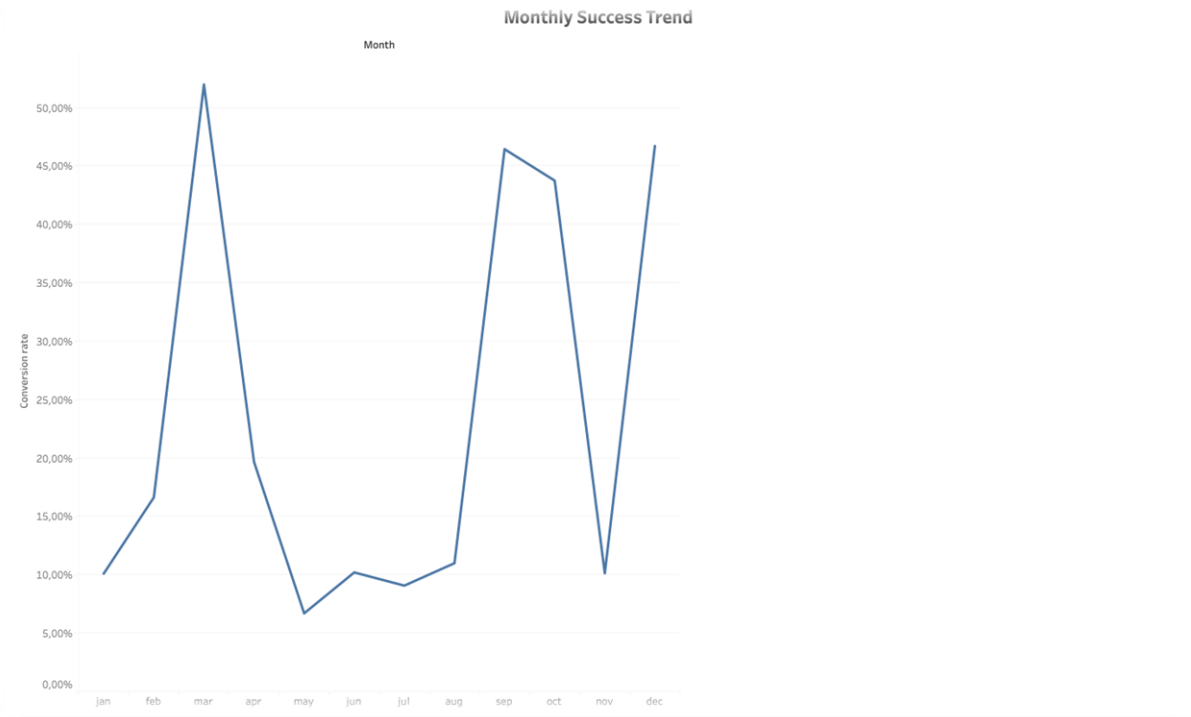


Figure 4 Note: Conversion rates exhibit high volatility throughout the year, with major performance spikes occurring in March, September, and October, contrasting with a notable decline in May

- **Peaks:** Significant spikes in conversion occur in **March, September, and October**.
- **Dips:** May is the lowest-performing month for conversion.
- **Tactical Advice:** Launch "Primary Campaigns" in Q1 and late Q3. Use low-conversion months (like May) for data cleaning and strategy rather than high-volume calling.

6. Conclusion & Next Steps

1. **Segmented Targeting:** Build a dedicated marketing list for students and retirees.
2. **Script Optimization:** Redesign call scripts to front-load value to beat the 4-minute drop-off.
3. **Budget Reallocation:** Shift 20% of the calling budget from May to the high-performing month of September.